

Random Forest Paper Reproduction Report

Diabetes Prediction Using the BRFSS 2015 Dataset

Paper: Efficient Diabetes Prediction Using Random Forests and Minimal Health Indicators on the BRFSS Dataset

Year: 2025

Algorithm: Random Forest

Dataset: BRFSS 2015 Diabetes Health Indicators

1. Objective

The objective of this work was to reproduce the Random Forest experiments presented in the selected paper using the same BRFSS dataset and the same model settings. The reproduced results were then compared with the values reported by the authors.

2. Methodology

The dataset contained 70,692 records, 21 input features and one binary target variable, where 0 represents no diabetes and 1 represents diabetes or prediabetes. The data was split into 80% training data and 20% testing data. No scaling or normalization was applied.

The following Random Forest experiments were performed:

- Default Random Forest using standard Scikit-learn parameters.
- Balanced Random Forest using `class_weight="balanced"`.
- Tuned Random Forest using `n_estimators=200`, `max_depth=10` and `min_samples_split=10`.
- The tuned model was retrained using the top 16, 12, 8 and 4 most important features.

3. Results Obtained

Model	Features	Train Acc.	Test Acc.	Precision	Recall	F1-score	Gap
Default Random Forest	21	0.995102	0.737110	0.717210	0.780394	0.747469	0.257992
Balanced Random Forest	21	0.995102	0.738383	0.719298	0.779401	0.748145	0.256719
Tuned Random Forest - All 21 Features	21	0.771100	0.752175	0.730584	0.796709	0.762215	0.018925
Tuned Random Forest - Top 16 Features	16	0.772267	0.751609	0.730124	0.795999	0.761640	0.020658
Tuned Random Forest - Top 12 Features	12	0.769880	0.748639	0.726448	0.795290	0.759312	0.021241

Tuned Random Forest - Top 8 Features	8	0.765583	0.747083	0.725432	0.792737	0.757592	0.018500
Tuned Random Forest - Top 4 Features	4	0.750889	0.740788	0.719912	0.785785	0.751407	0.010101

4. Comparison with the Paper

Model	Paper Accuracy	My Accuracy	Accuracy Difference	Paper F1	My F1	F1 Difference
Default Random Forest	0.737	0.737110	+0.000110	0.747	0.747469	+0.000469
Balanced Random Forest	0.738	0.738383	+0.000383	0.748	0.748145	+0.000145
Tuned Random Forest - All 21 Features	0.752	0.752175	+0.000175	0.762	0.762215	+0.000215
Tuned Random Forest - Top 16 Features	0.749	0.751609	+0.002609	0.756	0.761640	+0.005640
Tuned Random Forest - Top 12 Features	0.748	0.748639	+0.000639	0.755	0.759312	+0.004312
Tuned Random Forest - Top 8 Features	0.745	0.747083	+0.002083	0.751	0.757592	+0.006592
Tuned Random Forest - Top 4 Features	0.741	0.740788	-0.000212	0.751	0.751407	+0.000407

5. Observations

- The tuned Random Forest using all 21 features achieved the best performance, with an accuracy of 0.752175 and an F1-score of 0.762215.
- The tuned 21-feature results match the detailed values reported in the paper exactly to six decimal places.
- The default and balanced models also produced results extremely close to the paper. The balanced model showed only a very small improvement because the dataset was already evenly balanced.
- Reducing the model to the top 16 features caused only a slight decline in performance. Its accuracy was 0.751609 and its F1-score was 0.761640.
- Performance gradually decreased as the number of features was reduced from 16 to 12, 8 and 4, which follows the same trend described in the paper.
- The default and balanced models had very large training-test gaps of about 0.258, indicating strong overfitting. The tuned model reduced this gap to about 0.019, showing much better generalization.

- The top 4-feature model achieved an F1-score of 0.751407, almost identical to the paper value of 0.751, showing that a small feature set retained much of the predictive power.

6. Conclusion

The paper was reproduced successfully. The results obtained from the implementation were either identical or extremely close to the published results. The tuned Random Forest with all 21 features performed best, while the reduced-feature experiments confirmed that most predictive performance could be preserved with fewer variables. Tuning also greatly reduced overfitting compared with the default and class-weighted models.

7. Note on Reproducibility

The paper did not clearly state every random seed used in its experiments. However, the tuned model results and confusion-matrix values were reproduced exactly, while the rounded values for all other experiments closely followed the values reported in the paper.